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# =====
# AIM:
# Demonstrate the Effect of Sampling on Object Visibility
# Google Colab
# =====

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload a road or vehicle image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Create Low-Resolution Image
# -----
small = cv2.resize(img, (80, 80), interpolation=cv2.INTER_AREA)

# Upscale back to original size
low_resolution = cv2.resize(
    small,
    (img.shape[1], img.shape[0]),
    interpolation=cv2.INTER_NEAREST
)

# -----
# Sharpen Enhanced Image
# -----
kernel = [[0,-1,0],
          [-1,5,-1],
          [0,-1,0]]

import numpy as np
kernel = np.array(kernel)

enhanced = cv2.filter2D(low_resolution, -1, kernel)

# -----
# Display Images
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_resolution)

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plt.title( Low Resolution (Poor Sampling) )
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()

# -----
# Result
# -----
print("\n===== SAMPLING ANALYSIS =====")
print("Original Resolution :", img.shape[1], "x", img.shape[0])
print("Reduced Sampling      : 80 x 80 pixels")
print("Image Formation       : Reduced object clarity")
print("Enhanced Image         : Improved sharpness")
print("Observation            : Low sampling decreases object visibility.")
print("Conclusion              : Higher sampling improves object detection.")
print("=====")

```

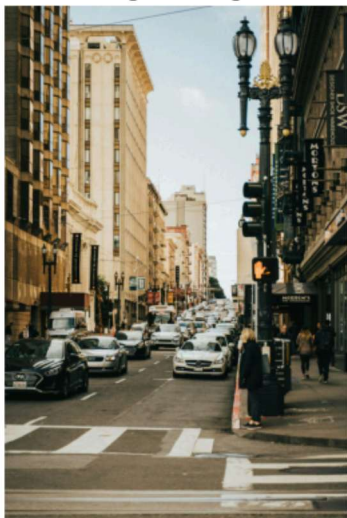
Upload a road or vehicle image

Q10.jpg

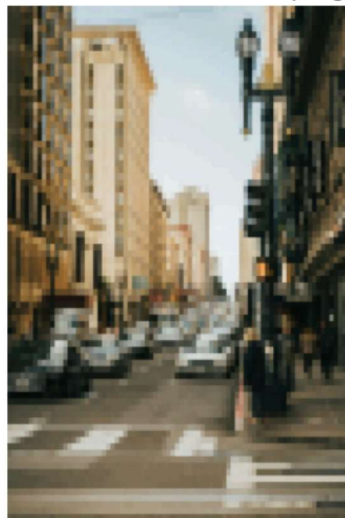
Q10.jpg(image/jpeg) - 2007749 bytes, last modified: 7/15/2026 - 100% done

Saving Q10.jpg to Q10.jpg

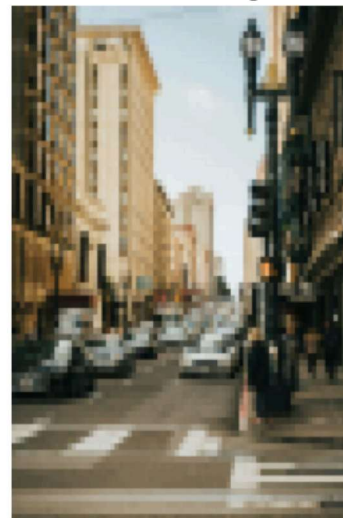
Original Image



Low Resolution (Poor Sampling)



Enhanced Image



===== SAMPLING ANALYSIS =====

Original Resolution : 3000 x 4500

Reduced Sampling : 80 x 80 pixels

Image Formation : Reduced object clarity

Enhanced Image : Improved sharpness

Observation : Low sampling decreases object visibility.

Conclusion : Higher sampling improves object detection.

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